

ANKITH REDDY AVULA

✉ ankithr005@gmail.com

🌐 [avulaankith.github.io](https://github.com/avulaankith)

🌐 [LinkedIn/avulaankith](https://www.linkedin.com/in/avulaankith)

☎ +1 (214) 675-8544

Education

University of Texas Arlington

August 2022 – May 2024

Master of Science in Computer Science GPA - 4.0

Arlington, TX

Courses: Cloud Computing and Big Data, Distributed Systems, Algorithm Design, Software Engineering, Distributed Systems, Database Systems, Object Oriented Programming, Operating Systems, Compilers, SDLC

IITDM Kurnool

August 2018 – May 2022

Bachelor of Technology in Computer Engineering GPA - 3.4

Kurnool, India

Courses: Data Structures, Database Systems, Object Oriented Programming, Operating Systems, Compilers

Technical Skills

Languages: C, C++, Java, Python, R, HTML, CSS, JavaScript, SQL, Scala, GoLang

Databases: SQL, MySQL, MongoDB, NoSQL, PostgreSQL

Big Data Tools: Apache (Hadoop, PySpark, Kafka, Cassandra), ETL, Scikit-learn, Pandas, NumPy, OpenCV, NLTK, Tableau, PowerBI, Matplotlib, Searborn, Snowflake

ML Tools: Scikit-learn, Pandas, NumPy, OpenCV, NLTK, Tableau, PowerBI, Matplotlib, Searborn, Snowflake

Machine Learning: Generative AI, Deep Learning, TensorFlow, Pytorch, Keras, LangChain

Web: React.js, Bootstrap, JQuery, Flask, Git, Node.js

Cloud: AWS (S3, EC2, RDS, Athena, Redshift, Glue, IAM, DynamoDB, Lambda, Kinesis, SageMaker), Docker, Kubernetes Azure (Databricks, Data Lake, SQL, VS Code, GitHub),

Experience

Astrosoft Technologies

January 2025 – Present

Machine Learning Engineer

Remote

- Designed and fine-tuned Large Language Models (**LLMs**) for domain-specific tasks, optimizing performance using **GPT**, **BERT**, and **LLaMA** architectures.
- Developed **RAG**-based AI systems, integrating **LangChain** and **vector databases** to enhance search and retrieval capabilities.
- Developed efficient fine-tuning strategies, implementing **LoRA** (Low-Rank Adaptation), **PEFT** (Parameter-Efficient Fine-Tuning), and **QLoRA** to enable faster model training with lower computational cost.
- Built scalable **LLM** inference pipelines leveraging **PyTorch**, **TensorFlow**, and Hugging Face **Transformers** to support real-time applications.
- Engineered low-latency microservices for AI model deployment using **Flask**, and **Node.js**, improving **API** response times by 30%.
- Designed and optimized distributed AI model training on cloud environments (**AWS**, **Azure**) utilizing **SageMaker** and **GPU** acceleration.
- Implemented **MLOps** pipelines using **GitLab CI/CD**, **Kubeflow**, and **MLflow**, ensuring efficient model retraining and versioning.
- Applied transfer learning and model distillation to improve efficiency, reducing model inference time by 40%.
- Developed custom monitoring solutions using **AWS CloudWatch** and ELK Stack, enhancing system observability and reducing incident response time by 40%.
- Worked with DevOps and Platform Engineers to ensure seamless deployment and infrastructure automation, reducing system downtime.

University of Texas Arlington

September 2024 – December 2024

Research Assistant

Arlington, TX

- Spearheaded the development of a sophisticated Retrieval-Augmented Generation (**RAG**) system designed to enhance the search and retrieval of complex college documentation, including academic records, research publications, and administrative files with PostgreSQL and API Gateway.
- Integrated **LangChain** with multiple college databases and content management systems, facilitating seamless and secure document access for faculty, administrative staff, and students.
- Implemented advanced natural language processing techniques to improve semantic search capabilities, enabling more accurate and context-aware document retrieval.
- Optimized the **RAG** system by fine-tuning large language models and embedding techniques, achieving a 40% reduction in document retrieval time and a 35% increase in search relevance accuracy.

Samsung

May 2021 – November 2021

Research Intern

Remote

- Developed an audio source separation model to isolate sound categories (vocals, music, drums, etc) from mixed audio tracks using **Python**, **TensorFlow**, **UNets**, **Auto-Encoders**, and **Librosa**
- Implemented Fourier Transforms to convert audio signals into spectrograms for frequency domain analysis and custom UNet-based extraction.
- Optimized model architecture, hyperparameters, and performance, achieving accurate separation of sound sources across diverse audio inputs.

Ismriti

June 2019 – July 2019

Data Science Intern

Kanpur, India

- Developed a real-time facial emotion recognition system that recognizes and classifies the live facial emotion of the user using **Python**, **CNN**, **TensorFlow**, and **OpenCV**
- Solved the data scarcity and class imbalance problem by augmenting the data, dataset creation and increasing the size of samples which have very less representation among other classes to make the model unbiased.
- Integrated the model with OpenCV for dynamic, real-time emotion detection, optimizing for real-world scenarios with improved performance and precision.

Projects

DYNAMIC KNOWLEDGE RETRIEVAL SYSTEM USING RAG | *Python*

- Developed a Retrieval-Augmented Generation (RAG) model using Hugging Face Transformers and LangChain, enabling real-time knowledge retrieval from unstructured documents.
- Integrated the system with OpenAI's GPT models to generate concise summaries and contextual responses, achieving a 45% improvement in information accuracy.

ELECTRICAL SUBSTATION SEGMENTATION | *Python, Pytorch, Attention UNet*

- Ranked **Top 10 in IEEE-ICETCI 2021** Competition organized by **ISRO** on 'Machine learning-based feature extraction of Electrical Substations from Satellite data'
- Developed an Attention UNet for the semantic segmentation of large remotely sensed images to extract small objects like electrical substations.
- Enhanced model precision by retraining images with mIoU below a certain threshold (τ), expanding the training dataset with less confident images.

TWITTER SENTIMENT ANALYSIS USING DEEP LEARNING | *Python, Pytorch, Tensorflow, BERT*

 [Github](#)

- Implemented feature-based learning with BERT, including CNN, LSTM, and BiLSTM, for sentiment analysis on Twitter data and explored different combinations to predict sentiments (positive, negative, neutral, or irrelevant) associated with Twitter entities.
- Handled sentiment analysis dataset, recognizing "irrelevant" as a distinct category, Collaborated on Jupyter Notebooks with team for testing and experimentation on models.

AI-POWERED RESUME MATCHER | *Python, Streamlit, Google Generative AI, LLM*

- Built a Streamlit-based web app using Google's Generative AI to match resumes against job descriptions, providing personalized feedback and keyword suggestions.
- Integrated with ATS systems for real-time resume parsing and evaluation, enhancing job seeker experience.

BLOGGING WEBSITE | *JavaScript, PHP, CSS, HTML, Bootstrap, MySQL, Software Development*

 [Github](#)

- Led the web application development of a full-stack blogging website using HTML, CSS, JavaScript as front-end, back-end as PHP, and MySQL, with features like user authentication, message posting, and replies.
- Managed deployment on a local server with XAMPP, enabling Apache and MySQL server configuration. Collaborated with a team to build a feature-rich blogging platform with robust user interaction capabilities.

NEIGHBOURHOOD ANALYSIS USING PYTHON | *Python, Folium, Foursquare API, Geocoder*

 [Github](#)

- Analyzed New York City neighborhoods by utilizing datasets and Foursquare API to identify areas with Indian cuisine restaurants, assessing their distribution and popularity.
- Performed data cleaning and visualization by merging location data with restaurant ratings and tips, using Folium to create choropleth maps highlighting key areas with high user ratings and potential for new Indian restaurants.

VISUAL TRANSFORMER FOR MEDICAL IMAGE ANALYSIS | *Python*

- Designed a Visual Transformer-based architecture for tumor detection in MRI scans, improving diagnostic accuracy by 30%.
- Leveraged Azure ML for scalable model training and deployment, integrating with PowerBI for real-time visualization.

TWITTER SENTIMENT ANALYSIS USING DEEP LEARNING | *Python, Pytorch, Tensorflow, BERT*

- Developed a BERT-based sentiment analysis model, achieving 85% accuracy in classifying Twitter sentiments. Optimized deep learning architectures (CNN, LSTM, BiLSTM) for improved prediction performance.
- Handled sentiment analysis dataset, recognizing "irrelevant" as a distinct category, Collaborated on Jupyter Notebooks with team for testing and experimentation on models.

HIGHWAY LANE SEGMENTATION with Visual Transformers | *Pytorch, Pytorch, Transformers*

- Implemented a Visual Transformer model for accurate lane detection in autonomous driving systems, surpassing traditional CNN performance.
- Integrated model outputs with real-time dashboards using Tableau, facilitating easy monitoring and analysis.

TWITTER DATA PIPELINE USING AIRFLOW | *Python, Spark, Docker, Apache Airflow, AWS, Pandas, ETL*

- Developed and maintained a robust data pipeline for ingesting real-time Twitter data, utilizing Twitter API to collect tweets and related metadata, ensuring reliable and continuous data flow for downstream analysis.
- Utilized Apache Airflow to orchestrate and schedule complex ETL workflows, automating the extraction, transformation, and loading (ETL) of Twitter data into a centralized database, enhancing data availability and operational efficiency.
- Employed data processing frameworks to clean, transform, and enrich Twitter data, storing the processed data in a PostgreSQL database.

LARGE SCALE LINEAR REGRESSION USING SPARK | *Scala, Apache Spark, Databricks*

- Implemented closed-form solution for large datasets using distributed matrices, leveraging Spark RDDs and Breeze library for matrix computations.
- Applied the outer-product technique to compute theta for enhanced performance and scalability, utilizing both Scala and DataFrame approaches.
- Executed the algorithm on the Boston Housing Dataset, demonstrating its efficacy in predicting housing prices with substantial improvements in computational efficiency.